COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE212 / Automating Server Management**	**SECTION:**	**CPE31S2**
DATE PERFORMED:	**September 30 2026**	**DATE SUBMITTED:**	**October 6 2026**
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ACTIVITY NO. 10.1 **Install, Configure, and Manage Log Monitoring tools**

1. PROCEDURE OUTPUT

1. Create a playbook that:
a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

Step 1 Creating directory and inventory file for HOA10.1

```
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA10.1$ tree
.
└── inventory.yaml

1 directory, 1 file
```

kenzo@workstation: ~/ansible_proj/CPE212_Lim/HOA10.1

```
  GNU nano 6.2                        inventory.yaml
all:
  children:
    Ubuntu:
      hosts:
        server1:
          ansible_host: kenzo@192.168.56.107

    CentOS:
      hosts:
        server2:
          ansible_host: kenzo@192.168.56.106
```

To start, we create a directory inside the CPE212 directory using mkdir HOA10.1 and nano inventory.yaml.

Step 2 Creating roles

```
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA10.1$ mkdir -p roles/elasticsearch/tasks
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA10.1$ mkdir -p roles/logstash/tasks
kenzo@workstation:~/ansible_proj/CPE212_Lim/HOA10.1$ mkdir -p roles/kibana/tasks
```

Here we created directories for the roles using mkdir -p and the respective names for each (elastic search, logstash, and kibana)

Step 3 Building install_logmonitor.yaml file

kenzo@workstation: ~/ansible_proj/CPE212_Lim/HOA10.1

GNU nano 6.2 install_logmonitor.yaml

```
---
- name: Install and configure Elasticsearch
  hosts: all
  become: true
  roles:
    - elasticsearch

- name: Install and configure Logstash
  hosts: all
  become: true
  roles:
    - logstash

- name: Install and configure Kibana
  hosts: all
  become: true
  roles:
    - kibana
```

Here we create a yaml file that contains the compilation of the roles we made earlier which are elasticsearch, logstash, and kibana. Due to this activity being done asynchronously in our personal computers, as a substitute of not being able to install on three servers, I modified that it would download each monitoring tool on both of my Server 1 and Server 2 (for context, my personal computer cannot handle 4 VMs at once)

Step 4 Building role/elasticsearch/task/main.yaml

kenzo@workstation: ~/ansible_proj/CPE212_Lim/HOA10.1/roles/elasticsearch/tasks

GNU nano 6.2 main.yaml

```
# 1 PREREQUISITE PACKAGE
- name: Install prerequisite packages (Ubuntu)
  apt:
    name: [apt-transport-https, curl, gnupg]
    state: present
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Install prerequisite packages (CentOS)
  yum:
    name: [curl, gnupg2]
    state: present
  when: ansible_distribution == "CentOS"

# 2 ELASTIC SIGNING KEY

- name: Add the Elastic GPG key (Ubuntu)
  shell: wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share>
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg
  when: ansible_distribution == "Ubuntu"

- name: Add the Elastic GPG key (CentOS)
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_distribution == "CentOS"
```

```
</share/keyrings/elasticsearch-keyring.gpg
```

The first part gets the tools apt and yum needs to download and verify Elastic's packages, and only runs on Ubuntu/CentOS respectively.

Ubuntu downloads Elastic's signing key so apt can confirm the packages are genuine.
CentOS also imports the same key into RPM's keyring so yum can verify the packages.

```
# 3 REGISTER THE REPOSITORY
# Ubuntu
- name: Add the Elastic 8.x APT repository (Ubuntu)
  copy:
    dest: /etc/apt/sources.list.d/elastic-8.x.list
    content: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elasti>
  when: ansible_distribution == "Ubuntu"

- name: Update apt cache after adding the repository
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

# CentOS
- name: Add the Elastic 8.x YUM repository (CentOS)
  yum_repository:
    name: elastic-8.x
    description: Elastic 8.x repository
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_distribution == "CentOS"
```

```
<elastic.co/packages/8.x/apt stable main\n"
```

This tells apt where to find Elasticsearch, then refreshes the package list so it picks up the new source.

```
# 4 INSTALL PACKAGES

- name: Install Elasticsearch (Ubuntu)
  apt:
    name: elasticsearch
    state: present
  when: ansible_distribution == "Ubuntu"

- name: Install Elasticsearch (CentOS)
  yum:
    name: elasticsearch
    state: present
  when: ansible_distribution == "CentOS"
```

This installs Elasticsearch using whichever package manager matches the machine.

Step 5 Building kibana role/task/main.yaml

kenzo@workstation: ~/ansible_proj/CPE212_Lim/HOA10.1/roles/kibana/tasks

GNU nano 6.2 main.yaml *

```
# PREREQUISITE PACKAGE
- name: Install prerequisite packages (Ubuntu)
  apt:
    name: [apt-transport-https, curl, gnupg]
    state: present
    update_cache: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Install prerequisite packages (CentOS)
  yum:
    name: [curl, gnupg2]
    state: present
  when: ansible_facts['os_family'] == "RedHat"

# ELASTIC SIGNING KEY
- name: Add the Elastic GPG key (Ubuntu)
  shell: wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share>
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg
  when: ansible_facts['os_family'] == "Debian"

- name: Add the Elastic GPG key (CentOS)
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_facts['os_family'] == "RedHat"
```

```
</share/keyrings/elasticsearch-keyring.gpg
```

```
# REGISTER THE REPOSITORY
- name: Add the Elastic 8.x APT repository (Ubuntu)
  copy:
    dest: /etc/apt/sources.list.d/elastic-8.x.list
    content: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elasti>
  when: ansible_facts['os_family'] == "Debian"

- name: Update apt cache after adding the repository
  apt:
    update_cache: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Add the Elastic 8.x YUM repository (CentOS)
  yum_repository:
    name: elastic-8.x
    description: Elastic 8.x repository
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_facts['os_family'] == "RedHat"
```

```
<elastic.co/packages/8.x/apt stable main\n"
```

Repeat the exact same pattern as Elasticsearch (prerequisites → GPG key → repository → install), just swapping the package name to kibana.

```
# INSTALL PACKAGES
- name: Install Kibana (Ubuntu)
  apt:
    name: kibana
    state: present
  when: ansible_facts['os_family'] == "Debian"

- name: Install Kibana (CentOS)
  yum:
    name: kibana
    state: present
  when: ansible_facts['os_family'] == "RedHat"
```

This installs the Kibana package.

Step 6 Building logstash role/tasks/main.yaml

kenzo@workstation: ~/ansible_proj/CPE212_Lim/HOA10.1/roles/logstash/tasks

```
 GNU nano 6.2                                    main.yaml *
# PREREQUISITE PACKAGE
- name: Install prerequisite packages (Ubuntu)
  apt:
    name: [apt-transport-https, curl, gnupg]
    state: present
    update_cache: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Install prerequisite packages (CentOS)
  yum:
    name: [curl, gnupg2]
    state: present
  when: ansible_facts['os_family'] == "RedHat"

# ELASTIC SIGNING KEY
- name: Add the Elastic GPG key (Ubuntu)
  shell: wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share>
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg
  when: ansible_facts['os_family'] == "Debian"

- name: Add the Elastic GPG key (CentOS)
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_facts['os_family'] == "RedHat"
```

```
</share/keyrings/elasticsearch-keyring.gpg
```

```
# REGISTER THE REPOSITORY
- name: Add the Elastic 8.x APT repository (Ubuntu)
  copy:
    dest: /etc/apt/sources.list.d/elastic-8.x.list
    content: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elas
  when: ansible_facts['os_family'] == "Debian"

- name: Update apt cache after adding the repository
  apt:
    update_cache: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Add the Elastic 8.x YUM repository (CentOS)
  yum_repository:
    name: elastic-8.x
    description: Elastic 8.x repository
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_facts['os_family'] == "RedHat"
```

```
<elastic.co/packages/8.x/apt stable main\n"
```

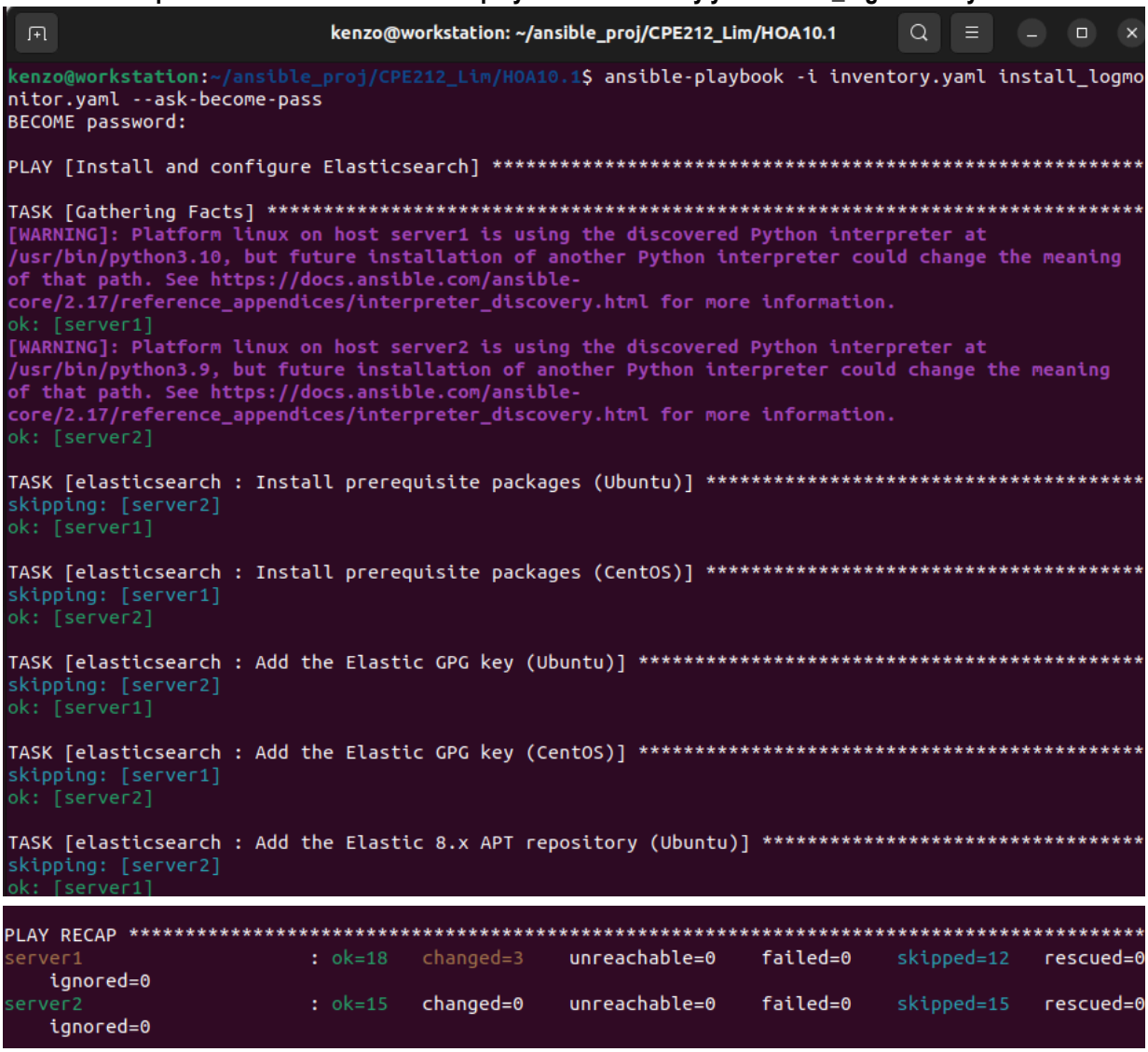
Repeat the exact same pattern as Elasticsearch (prerequisites → GPG key → repository → install), just swapping the package name to logstash.

```
# INSTALL LOGSTASH
- name: Install Logstash (Ubuntu)
  apt:
    name: logstash
    state: present
  when: ansible_facts['os_family'] == "Debian"

- name: Install Logstash (CentOS)
  yum:
    name: logstash
    state: present
  when: ansible_facts['os_family'] == "RedHat"
```

This installs the logstash package.

Step 7 Run the command ansible-playbook -i inventory.yaml install_logmonitor.yaml



As we can see in the result, it was a successful installation for both Server 1 and Server 2.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What are the benefits of having log monitoring tool?

A centralized log monitoring tool makes it easier to monitor multiple servers in one place, quickly identify errors and system problems, and detect suspicious activity through collected security and access logs. It also reduces the need to manually check each server by automatically gathering logs and sending alerts when issues are detected, making system management faster and more efficient.

Conclusion:

Due to hardware limitations of my personal computer being unable to run four virtual machines simultaneously, I modified the install_logmonitor.yaml by installing the three monitoring tools (elasticsearch, kibana and logstash included) across my existing Server 1 and Server 2 nodes. Through this activity, I learned that adding GPG keys is important before package installation because it cryptographically verifies package authenticity, ensures file integrity, and prevents untrusted software repositories from compromising the system. Although I only finished installing the packages and was not able to configure the monitoring tools each, It was still a valuable knowledge acquired and experience gained from it.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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